

# Constraint Encoding in Physics-Informed Neural Networks for PDEs

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## 1 Introduction

### 1.1 Machine Learning and PDEs

Machine learning provides a flexible framework for approximating high-dimensional functions by composing affine maps with nonlinear activation functions and fitting the resulting parameters through loss minimization. In the present setting, the relevant point is not machine learning in full generality, but the fact that differentiable neural networks can be used as trial functions whose outputs, derivatives, and parameter dependence are all accessible during training. This makes them natural candidates for approximating solutions of partial differential equations, where one wants to represent an unknown field  $u$ , evaluate how well it satisfies a governing equation, and optimize over a rich function class. [1, 2]

The connection between machine learning and PDEs first developed largely through data-driven strategies. In one common approach, a neural network is trained on solution fields produced by a numerical solver and learns a map from problem inputs to the corresponding PDE solution. Closely related variants learn time-stepping rules for evolution equations or approximate full solution operators over families of coefficients, forcings, and initial data. These approaches can be highly effective, but they rely on numerical or observational data and therefore inherit the cost and limitations of the data source used for training. [2, 3]

Physics-informed methods shift the emphasis from data alone to direct encoding of analytical structure. Rather than only asking a neural network to match precomputed solutions, one can incorporate differential operators, boundary traces, initial conditions, weak formulations, variational principles, or admissibility constraints into the training procedure itself. This allows for generated solutions to be much more physically descriptive, as well as being much more accurate without requiring sampled or numerical data. [1, 3]

### 1.2 Data-Driven Neural PDE Solvers

Before physics-informed methods became a major direction, neural approaches to partial differential equations were often developed in a purely data-driven setting. In that framework, one first generates or collects solution data, typically from a trusted numerical solver, and then trains a network to approximate the map from problem inputs to the corresponding solution. Depending on the setting, this may mean learning a direct field-to-field regression map on a fixed grid, a time-stepping rule for an evolution equation, or an operator that acts on functions such as coefficients, forcings, or initial data and returns the associated PDE solution. [2, 5]

These data-driven strategies are attractive because, once trained, they can evaluate very quickly and may provide efficient surrogates for repeated simulation. At the same time, they are fundamentally tied to the data used in training. Their success depends on the representativeness and quality of the numerical or observational samples, and the learned map is most reliable on the distribution of problems that those samples actually cover. In this sense, the network is learning from examples of solutions rather than from the analytical structure of the PDE itself. [3]

This distinction is important to understand what changes once the governing equation, boundary data, initial data, or related analytical properties are built directly into the learning procedure. That shift leads naturally to physics-informed neural networks and related scientific machine learning methods, where the central object is no longer only a dataset of solution pairs, but the PDE structure itself.

### 1.3 Physics-Informed Neural Networks

Physics-informed neural networks replace or supplement dense solution supervision by enforcing the PDE through the training objective and, in some cases, through the architecture of the trial function itself. A neural network is used as an approximation  $u_\theta$  to the unknown solution, and the loss is constructed from quantities such as the

interior differential residual, boundary and initial condition mismatch, and possibly additional terms coming from observations, weak formulations, variational principles, or qualitative constraints. In contrast with purely data-driven solvers, the network is therefore trained not only to fit examples, but to satisfy mathematical structure drawn from the PDE. [1, 2, 3]

At first this may seem like a simple idea, just adding the differential residual to the loss and optimizing. In practice, however, the phrase “physics-informed” covers several genuinely different mechanisms. Boundary and initial conditions may be imposed softly through penalties or hard-wired into the admissible trial space. The PDE may be enforced in strong form at collocation points, in weak form against test functions, or through a variational functional when an energy structure is available. Additional penalties may also be introduced to encourage comparison-principle behavior, bounds, or entropy admissibility. These choices are not merely implementation details. They determine what kind of mathematical object the network is actually being asked to approximate and what type of correctness its training objective is capable of measuring. [6, 7, 8, 9, 3]

This becomes especially important once different PDE classes are compared. For coercive elliptic equations, weak and variational formulations are often the natural analytical setting, and admissible boundary traces as well as minimum and maximum principles play a central role. For parabolic problems, one must distinguish interior evolution from the initial surface and the spatial boundary, and can be further bounded by super and subsolution sets. For hyperbolic conservation laws, the issue is sharper still: after shock formation, weak solutions need not be unique, and the relevant target is the entropy solution rather than an arbitrary function with a small sampled residual. The motivation for what follows is therefore not simply to catalogue PINN variants, but to examine how different forms of constraint encoding align with different notions of solution and different notions of correctness across these PDE classes. [10, 11, 12, 13, 3]

## 2 Background: Neural PDE Solvers and Constraint Encoding

### 2.1 Neural Network Architectures as Differentiable Function Approximators

A feedforward neural network is a parameterized map obtained by composing affine transformations with nonlinear activation functions. If the input is denoted by  $z$ , a network with parameters  $\theta$  produces an output  $u_\theta(z)$  by repeatedly applying matrix-vector multiplication, bias addition, and activation at each layer. From the point of view of partial differential equations, the important feature is not only that such networks are flexible approximators, but that they are differentiable with respect to both their inputs and their parameters. This makes them usable as trial functions whose derivatives can be evaluated during training, allowing the network to be compared directly against differential operators, boundary conditions, and other PDE constraints. [14, 1]

The most basic architecture used in deep learning is the multilayer perceptron, or fully connected feedforward network. These networks treat the solution as a function of coordinates and represent it by repeated nonlinear transformations of those coordinates. This makes them especially natural for physics-informed formulations, where one wants direct access to derivatives with respect to the input variables. [14, 1]

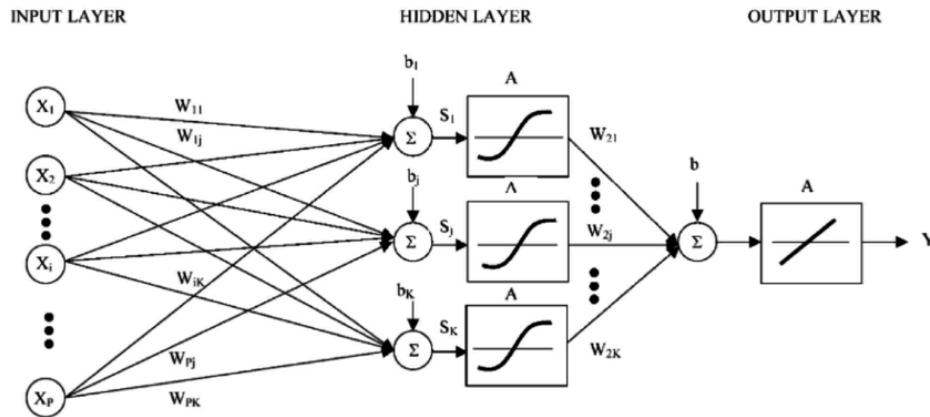


Figure 1: Basic neural-network background for the present work: a feedforward architecture mapping coordinates to a scalar field value, the role of differentiable activations, and the use of loss minimization to fit the network parameters.

Activation functions play a particularly important role in neural PDE solvers because derivatives of the network output enter the loss explicitly. In ordinary regression tasks, activations are often discussed mainly in terms of expressivity and trainability. In the present setting, they also determine whether higher derivatives are well behaved and numerically useful. Smooth activations such as tanh are therefore common in PINN-type models, especially when second derivatives appear in the governing equation. Depth and width also matter, but not only through approximation power: they affect optimization stability, conditioning, and the ease with which different scales in the solution can be represented. [14, 3]

Training is carried out by minimizing a loss function over the network parameters. In standard machine learning this loss measures mismatch between predictions and target labels. In neural PDE solvers the loss may instead involve differential residuals, boundary mismatch, initial mismatch, variational energies, or other physics-informed quantities. This change in the meaning of the loss is one of the central transitions from ordinary deep learning to scientific machine learning for PDEs. [2]

Some architectural and optimization choices are also necessary to consider. Layer width, network depth, activation smoothness, learning rate, sampling strategy, and number of training epochs all affect whether the chosen formulation can actually be trained successfully. In particular, composite losses may become imbalanced, and residual-based training may emphasize some constraints far more strongly than others.[14] For that reason, the computational section of this report keeps many low-level training choices fixed across experiments and varies primarily the PDE-informed ingredients themselves. This makes it easier to interpret differences between models as consequences of constraint encoding rather than as artifacts of unrelated hyperparameter choices. The resulting comparisons are therefore aimed not at producing the single best neural architecture, but at clarifying how different ways of incorporating PDE structure change the mathematical meaning and practical behavior of the learned solution. [2, 3]

## 2.2 Input/output Structure for Neural PDE Solvers

The input and output of a neural PDE solver depend on the formulation being used. In the coordinate-based setting adopted here, the input is a point in the independent-variable domain and the output is the predicted value of the solution at that point. Thus an elliptic problem on a two-dimensional domain uses a network of the form

$$u_\theta : (x, y) \mapsto \mathbb{R},$$

while parabolic and hyperbolic problems use

$$u_\theta : (x, t) \mapsto \mathbb{R}.$$

This pointwise representation is simple, but it is also what makes automatic differentiation and direct residual evaluation possible. [1, 2]

More broadly, neural PDE methods can also be formulated with different input-output structures. A purely data-driven model may take in coefficients, forcings, or initial data and output an entire solution field or a time-evolution map, while neural operator methods aim to learn mappings between function spaces rather than only pointwise coordinate-to-value relations. [3]

## 2.3 Soft Constraints, Hard Constraints, and Reformulations

Once the PDE itself is brought into the learning problem, several different mechanisms become possible. The most familiar is soft enforcement through penalty terms in the loss. A boundary condition, for example, may be included by sampling points on the boundary and penalizing mismatch there. Likewise, an initial condition or a positivity requirement may be encouraged by adding corresponding terms to the objective. [1, 2]

A second mechanism is hard enforcement, where the trial space is designed so that a constraint is satisfied identically. For Dirichlet problems, one can multiply the free neural component by a factor that vanishes on the boundary and add an extension of the prescribed trace. This changes the admissible class itself rather than merely penalizing violations after the fact. In practice, this distinction is mathematically important: a soft constraint asks the optimizer to trade off competing objectives, whereas a hard constraint removes part of that tradeoff by construction. [8, 9]

A third mechanism is reformulation of the PDE problem. Instead of enforcing the strong differential equation pointwise, one may pass to a weak formulation, a variational principle, or an admissibility condition adapted to the PDE class being studied. For elliptic problems, weak and variational formulations are often the most natural. For parabolic problems, comparison-based qualitative structure may matter in addition to the residual. For hyperbolic conservation laws, admissibility through entropy conditions becomes essential after shock formation. [7, 3, 12]

### 3 PDE-Theoretic Background and Notions of Correctness

#### 3.1 Strong Residual Formulations

The most direct way to encode a partial differential equation in a neural model is through its strong form. If the governing equation is written abstractly as

$$\mathcal{N}[u] = f,$$

then a neural approximation  $u_\theta$  induces the pointwise residual

$$R_\theta = \mathcal{N}[u_\theta] - f.$$

A strong-residual method penalizes this quantity at sampled points in the domain, typically through a loss of the form

$$\mathcal{L}_{\text{res}} = \frac{1}{N} \sum_{j=1}^N |R_\theta(z_j)|^2.$$

This is the basic mechanism underlying the original PINN formulation and remains the most common entry point into physics-informed learning. [1, 2]

From a PDE point of view, however, strong residual minimization should be interpreted carefully. The PDE is not being enforced as an identity on the full domain, but only through an empirical approximation of a continuum quantity. The sampled residual therefore measures one notion of correctness, but not necessarily the most natural one for every PDE class. For smooth elliptic or parabolic problems it may be a reasonable starting point, while for hyperbolic problems with shocks it can fail to distinguish between different weak continuations of the same initial data. This distinction between sampled strong residual and the natural PDE solution concept will be central in the sequel. [3, 12]

#### 3.2 Hard Enforcement and Admissible Trial Spaces

A different strategy is to restrict the trial space itself so that certain constraints are satisfied identically. In classical PDE language, one often seeks a solution in an admissible class determined by the boundary or initial conditions. Neural methods can mimic this by using an ansatz of the form

$$u_\theta = g^* + \phi N_\theta,$$

where  $g^*$  extends prescribed data and  $\phi$  vanishes on the relevant constraint set. Then every admissible network output satisfies the trace condition by construction. [8]

This is more than a numerical convenience. A soft penalty merely discourages violation of a constraint, while hard enforcement changes the class over which optimization takes place. For this reason, hard enforcement is especially natural when boundary or initial traces are readily provided, as in elliptic Dirichlet problems or time-dependent equations with distinguished initial surfaces. [11, 10]

#### 3.3 Weak Formulations

Many PDEs are most naturally understood not in pointwise strong form, but in weak form. Rather than requiring the derivatives in  $\mathcal{N}[u] = f$  to exist classically, one asks that the equation hold after integration against test functions. For elliptic problems, this is the setting in which Sobolev spaces, traces, and bilinear forms arise naturally. If  $u$  is sought in a space such as  $H^1(\Omega)$  or  $H_0^1(\Omega)$ , then the weak formulation often becomes the correct analytical notion of solution even when classical differentiability is unavailable. [10, 11]

This viewpoint matters for neural PDE solvers because a weak formulation changes both the derivative requirements on the trial function and the meaning of correctness. A model trained to satisfy a weak identity is not simply a softened version of a pointwise residual method; it is targeting a different mathematical statement. Variational PINNs and related methods exploit this idea by replacing collocation in the strong form with integral identities against test functions, often using integration by parts to lower the derivative order appearing on the neural ansatz. [6, 7]

### 3.4 Weak Solutions for Conservation Laws

Hyperbolic conservation laws differ sharply from elliptic and parabolic equations because discontinuities can develop from smooth or piecewise regular initial data. For a scalar conservation law

$$u_t + f(u)_x = 0,$$

classical solutions may fail to exist after shock formation, and one must enlarge the notion of solution to admit discontinuities. This leads to the weak formulation, in which the equation is required to hold only after integration against smooth compactly supported test functions. [12]

The crucial point is that weak formulation alone is not enough. Once discontinuities are allowed, weak solutions are typically not unique. This is exactly what makes the hyperbolic setting conceptually different from the elliptic and parabolic cases considered here. A neural method that merely reduces a sampled strong residual does not automatically identify the physically relevant shock solution. The question is therefore no longer only how the PDE is enforced, but which weak solution concept is actually being approximated. [12, 3]

### 3.5 Entropy Solutions and Admissibility

To recover uniqueness in shock-forming hyperbolic problems, one imposes an additional admissibility condition, usually expressed through entropy inequalities. For each convex entropy  $\eta$  with corresponding entropy flux  $q$ , an entropy solution satisfies

$$\eta(u)_t + q(u)_x \leq 0$$

in the weak sense. This rules out nonphysical weak solutions and selects the physically admissible one. In Burgers' equation, the entropy condition distinguishes the correct compressive shock from other weak continuations that satisfy the conservation law formally but violate admissibility. [12]

In the hyperbolic case, entropy is therefore not an optional refinement but the central additional structure beyond the conservation law itself. The informed hyperbolic model augments the standard residual by an entropy-violation penalty, reflecting the fact that correctness for shock solutions must include admissibility. This is the most direct way in the present project to make the neural objective sensitive to the difference between an arbitrary weak solution and the entropy solution. [12]

### 3.6 Variational and Ritz Formulations

When a PDE is the Euler–Lagrange equation of an energy functional, one may solve it by minimizing that functional over an admissible space. This is the classical Ritz viewpoint. For coercive elliptic problems, variational structure is often more natural than pointwise residual minimization because it aligns directly with existence, uniqueness, and stability theory. In this setting, correctness is tied not only to residual smallness, but to whether the neural trial function lies near the minimizer of the appropriate energy. [11, 10]

This distinction is especially important for the elliptic class of PDEs. There, the informed model is not only boundary-admissible but also trained through a discrete Ritz functional rather than a standard strong residual. Conceptually, this means the method is approximating the variational problem itself, not merely sampling the differential operator at interior points. The change is therefore both numerical and analytical: it replaces one notion of enforcement by another more closely matched to the PDE class. [6, 3]

### 3.7 Maximum and Comparison Principles

Maximum and comparison principles provide qualitative control over solutions that is not captured by residual norms alone. In elliptic and parabolic settings, these principles say that extrema or orderings in the interior are constrained by boundary, initial, forcing, and coefficient data under suitable assumptions. The comparison principle is a more general statement: if  $u$  and  $v$  are ordered on the relevant boundary set and satisfy compatible differential inequalities, then the same ordering persists in the domain. Maximum and minimum principles arise as special cases when one compares a solution against constants or barrier functions. [13, 10, 11]

For neural PDE solvers, these principles are useful in two distinct ways. First, they motivate qualitative diagnostics: a learned solution may have low global error and yet violate a sign condition or leave an expected barrier region. Second, they suggest soft constraints built from subsolutions and supersolutions. In the parabolic case, this becomes a natural way to encode comparison-principle behavior without claiming that the full theorem has been hard-wired into the architecture. In the elliptic case, related barrier ideas can be used as secondary qualitative penalties alongside the main variational formulation. [13]

## 4 Experimental Framework

All code can be accessed at [github.com/zane-perry/pde-pinns](https://github.com/zane-perry/pde-pinns).

### 4.1 Computational Design

All computational comparisons in this report are built around a common baseline architecture so that differences between models can be attributed primarily to the PDE-informed modifications rather than to unrelated changes in network design. In each case, the baseline model is a fully connected feedforward network with smooth activation functions, trained through gradient-based optimization. The network takes the independent variables of the PDE as input and returns a scalar approximation  $u_\theta$  of the solution field. This coordinate-based representation is standard in PINN-type methods because it allows automatic differentiation with respect to spatial and temporal variables and therefore direct evaluation of differential operators in the loss. [1, 14, 2]

In addition to the informed models developed for each PDE class, the experiments also include a standard uninformed neural-network baseline. This baseline uses the same general feedforward architecture but is trained in the most naive physics-informed manner considered in the report: a strong residual objective together with soft penalty terms for the relevant boundary or initial conditions. It is called uninformed here not because it ignores the PDE entirely, but because it does not incorporate the PDE-specific structural modifications that motivate the rest of the study.

The informed models are constructed by modifying this common baseline as little as possible outside the PDE-specific mechanism under study. Thus the main variations across experiments come from how constraints are encoded: hard trace enforcement, variational training, qualitative barrier penalties, or entropy-aware admissibility terms. Optimizer family, epoch budget, and general training style are kept aligned across the models in order to support interpretable comparisons.

### 4.2 Benchmark Problems and Informed Models

This computational study uses one benchmark from each of the main three PDE classes. The parabolic benchmark is a one-dimensional advection–diffusion problem with a known exact solution, chosen so that interior evolution, spatial boundary conditions, and initial data all play visible and distinct roles. The hyperbolic benchmark is Burgers’ equation in a shock-forming Riemann regime, with the entropy solution taken explicitly as the target. The elliptic benchmark is a coercive variable-coefficient Dirichlet problem with a manufactured exact solution, chosen to support weak and variational interpretation as well as nontrivial boundary trace handling.

For each PDE class, the common uninformed model is compared against a single informed model designed around the main analytical idea of that section. In the parabolic case, the informed model enforces initial and boundary conditions exactly and supplements the strong residual with a soft barrier term derived from explicit subsolution and supersolution information. In the hyperbolic case, the informed model augments the conservation-law residual by an entropy-aware penalty intended to favor the admissible shock solution. In the elliptic case, the informed model uses a hard boundary-admissible ansatz together with a Ritz-type variational loss, supplemented by a soft max/min-style qualitative penalty.

Because the benchmark problems have known exact or target solutions, they also permit a direct comparison between physics-informed training objectives and actual solution error. This is useful not because exact solutions are expected in realistic applications, but because such benchmarks make it possible to separate questions of approximation quality from questions of interpretability and correctness.

### 4.3 Evaluation Metrics

The models are evaluated using both common and PDE-specific metrics. Across all three PDE classes, the most basic common quantities are relative solution error on a dense evaluation grid and held-out residual diagnostics. These quantities provide a first indication of whether the informed models improve on the uninformed baseline, but they are not treated as sufficient on their own. Accordingly, each case study also includes metrics adapted to its natural analytical structure. In the parabolic setting, this includes final-time error and diagnostics associated with the barrier-based qualitative constraints. In the hyperbolic setting, the most informative metrics include relative  $L^1$  error, shock location error, shock width, overshoot and undershoot near the discontinuity, and entropy-violation indicators. In the elliptic setting, boundary-trace error,  $H^1$ -type quantities, and variationally motivated diagnostics are more natural. [10, 11, 12]

## 5 Parabolic Problems

The parabolic case is a natural first step away from the uninformed baseline because it remains close to the standard strong-residual PINN framework while already exhibiting several features that are not captured well by a naive loss alone. A parabolic problem has an interior evolution equation, an initial surface, and spatial boundary traces that all play distinct analytical roles. This makes it a useful setting in which to study how the treatment of traces can be improved without yet changing the formulation as drastically as in the elliptic variational case or introducing the admissibility issues of the hyperbolic case. [11, 10, 1] The differential equation is still enforced in strong form through a sampled residual, but the informed model enriches that baseline in two ways: it enforces the initial and boundary conditions exactly through the trial space, and it adds a soft qualitative penalty based on explicit subsolution and supersolution information. [2, 3]

### 5.1 Parabolic Comparison and Maximum-Principle Viewpoint

Parabolic equations admit comparison principles that provide qualitative control beyond small residual error. In broad terms, if a subsolution lies below a supersolution on the parabolic boundary and both satisfy the appropriate differential inequalities, then that ordering persists throughout the space-time domain. Maximum and minimum principles arise as special cases of this viewpoint. These principles are central in the analysis of parabolic equations because they control sign, bounds, and ordering properties that are not visible from a single global norm alone. [13]

In this example, the comparison principle is not imposed as a theorem-level hard constraint. Instead, it motivates a soft qualitative penalty. The informed parabolic model is supplied with explicit subsolution and supersolution barriers, and the loss penalizes excursions outside this expected region. This does not prove a full comparison theorem for the neural approximation, but it does bias training toward solutions that respect the qualitative parabolic structure suggested by the PDE theory. [13]

### 5.2 Parabolic Benchmark Problem

The benchmark parabolic problem is a one-dimensional advection–diffusion equation on  $(x, t) \in [0, 1] \times [0, 1]$ ,

$$u_t - u_{xx} + 2u_x = f(x, t),$$

with exact solution

$$u(x, t) = e^{-t} \sin(\pi x) + t x(1 - x).$$

The forcing  $f$  is chosen accordingly so that this function satisfies the PDE exactly. The associated initial and boundary conditions are

$$u(x, 0) = \sin(\pi x), \quad u(0, t) = u(1, t) = 0.$$

This example is convenient because it has a closed-form solution, separates the roles of the initial condition and the spatial boundary, and admits simple explicit barriers,

$$\underline{u}(x, t) = e^{-t} \sin(\pi x), \quad \bar{u}(x, t) = e^{-t} \sin(\pi x) + 2t x(1 - x).$$

These provide a natural source of qualitative information for the informed loss.

### 5.3 Hard Enforcement of Initial and Boundary Conditions

To satisfy the trace conditions exactly, the informed model uses an ansatz of the form

$$u_\theta(x, t) = \sin(\pi x) + t x(1 - x) N_\theta(x, t),$$

where  $N_\theta$  is the free neural component. This construction is chosen so that when  $t = 0$ , the second term vanishes and one recovers

$$u_\theta(x, 0) = \sin(\pi x),$$

while for  $x = 0$  or  $x = 1$ , the multiplicative factor  $x(1 - x)$  vanishes and therefore

$$u_\theta(0, t) = u_\theta(1, t) = 0.$$

The initial and boundary conditions are therefore satisfied by construction for every choice of network parameters. This is analytically more meaningful than merely adding soft trace penalties to the loss. Instead of asking the optimizer to trade off PDE residual against trace mismatch, the admissible class is restricted from the start.

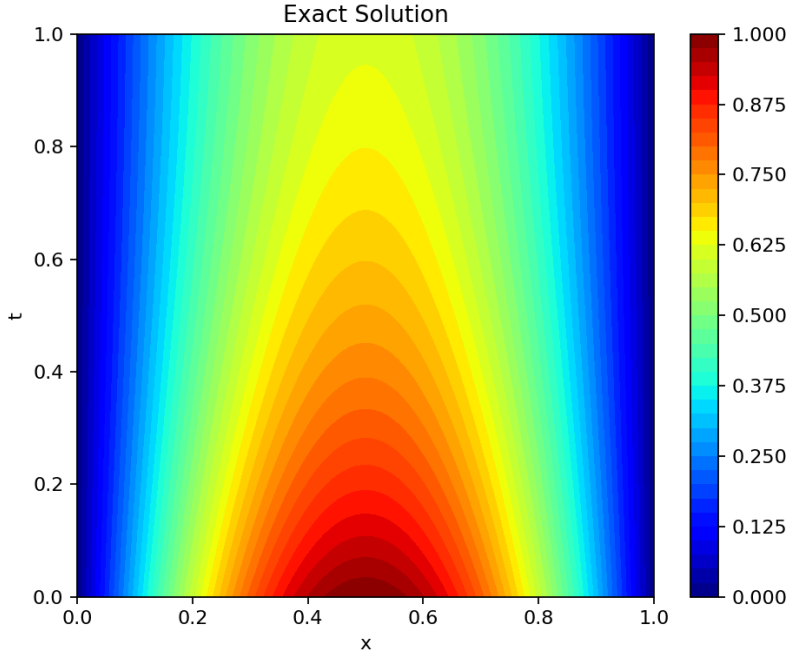


Figure 2: Exact solution for the parabolic benchmark problem, shown over the full space-time domain.

#### 5.4 Soft Subsolution/Supersolution Barrier Encoding

The second informed ingredient is a soft barrier loss based on the explicit functions  $\underline{u}$  and  $\bar{u}$ . Since these are chosen to act as lower and upper bounds for the target solution, the informed model is penalized when its prediction leaves the interval they define. At the loss level, this takes the form of a penalty on the positive parts of

$$\underline{u} - u_\theta \quad \text{and} \quad u_\theta - \bar{u}.$$

This encourages the learned solution to remain within a comparison-principle-inspired admissible region without requiring an exact hard encoding of the full theorem. [13]

#### 5.5 Numerical Results

The parabolic experiments show that these informed modifications produce a solution that is both quantitatively accurate and qualitatively well controlled. The informed model achieved a relative test error of approximately

$$1.39 \times 10^{-4},$$

with a held-out residual mean square error of approximately

$$7.62 \times 10^{-6}.$$

Since the initial and boundary conditions are built into the ansatz, the corresponding numerical trace errors are essentially at machine precision, with maximum initial-condition and boundary-condition discrepancies on the order of

$$10^{-7}.$$

The barrier diagnostics are similarly small. The final-time relative  $L^2$  error is approximately

$$2.50 \times 10^{-4},$$

and the mean barrier violation is approximately

$$5.85 \times 10^{-10}.$$



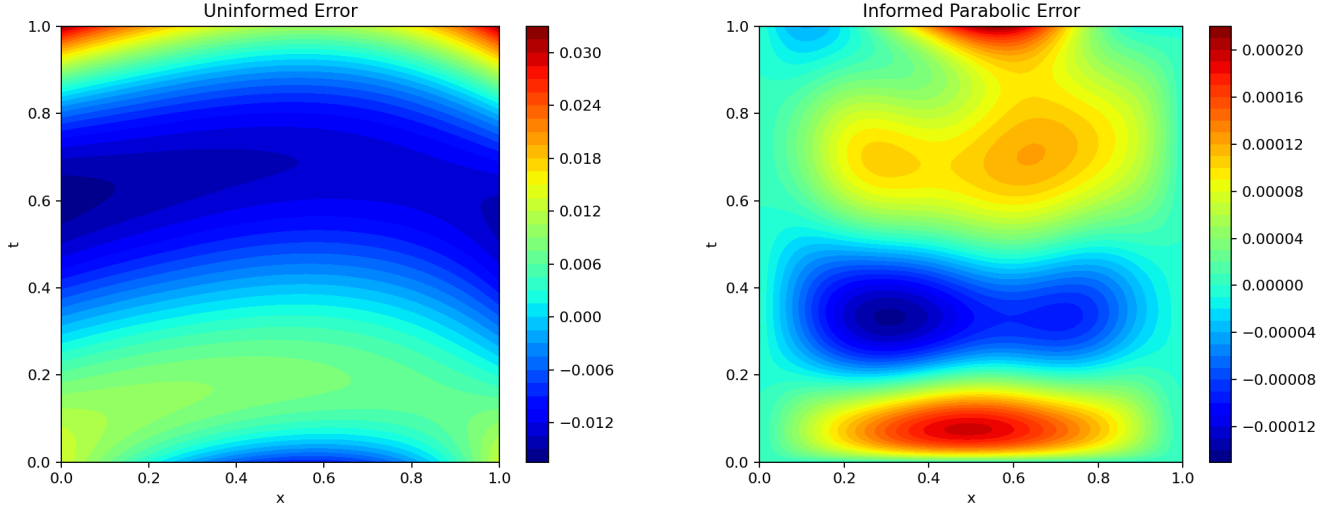


Figure 3: Error comparison for the parabolic benchmark. The uninformed model is shown on the left and the informed model is shown on the right. The informed model reduces the error by roughly two orders of magnitude and confines it to much smaller regions.

The maximum lower- and upper-barrier violations remain on the order of  $10^{-7}$ , indicating that the learned solution stays extremely close to the comparison-principle-inspired admissible region throughout the domain.

The error plots make this contrast especially clear. The uninformed model exhibits a broad, domain-wide error pattern with visible positive and negative bias across much of the space-time region. Its error is not confined to a small localized region, but instead varies smoothly across the domain. By contrast, the informed model has an error field that is dramatically smaller in magnitude and much more organized. Its error is shown as an absolute error plot and is concentrated in a few thin horizontal bands in time, with the strongest band occurring near early times and the remaining bands appearing much more weakly later in the evolution. This visual difference is important for interpretation. The uninformed model does not simply make a slightly larger error; rather, it exhibits a global bias pattern on the order of  $10^{-2}$ , while the informed model reduces the error to the order of  $10^{-4}$  and confines it to much smaller regions.

## 6 Hyperbolic problems

The hyperbolic setting is qualitatively different from the parabolic case because the main difficulty is no longer only how to enforce traces or qualitative barriers, but what notion of solution is being targeted after discontinuities form. For a conservation law such as Burgers' equation, the strong differential form is meaningful only as long as the solution remains smooth. Once shock formation occurs, the classical residual loses its status as a complete description of correctness, since the physically relevant solution is no longer characterized by pointwise differentiability. [12] This is why a naive strong-residual PINN becomes conceptually inadequate in the shock regime. A network may still reduce a sampled residual and yet fail to represent the correct discontinuous solution structure. In particular, a small strong residual does not by itself encode the admissibility conditions needed to distinguish physically meaningful shocks from other weak continuations. [2]

### 6.1 Weak Solutions and Nonuniqueness

For a scalar conservation law

$$u_t + f(u)_x = 0,$$

classical solutions may cease to exist after characteristics intersect. One must then pass to the weak formulation, in which the equation is required to hold only after integration against smooth compactly supported test functions. notion of solution allows discontinuities, and therefore makes shock solutions mathematically admissible. [12]

The difficulty is that weak solutions are generally not unique. That is the central analytical obstacle in the hyperbolic setting. In the hyperbolic shock regime, weak formulation alone is not sufficient to identify the correct physical solution. [12, 3]

## 6.2 Entropy solutions and admissibility

The additional selection principle is entropy admissibility. For Burgers' equation, the physically relevant shock solution is the entropy solution, characterized by the entropy inequality rather than by the conservation law alone. More generally, for each convex entropy  $\eta$  with corresponding entropy flux  $q$ , an admissible solution satisfies

$$\eta(u)_t + q(u)_x \leq 0$$

in the weak sense. This excludes nonphysical weak solutions and restores uniqueness in the appropriate solution class. [12]

In the computational setting of this report, entropy is incorporated through a soft loss term. The informed hyperbolic model retains the baseline conservation-law residual, but augments it by penalizing violations of a quadratic entropy inequality. This does not reproduce the full theory of entropy solutions, but it does make the optimization problem sensitive to admissibility in a way that the uninformed baseline is not. [12]

## 6.3 Benchmark Hyperbolic Problem

The hyperbolic benchmark is Burgers' equation

$$u_t + \left( \frac{u^2}{2} \right)_x = 0,$$

posed in a shock-forming Riemann regime. The initial data are

$$u(x, 0) = 2 \text{ for } x < 0, \quad u(x, 0) = 0 \text{ for } x > 0,$$

so that the entropy solution is a shock moving with Rankine–Hugoniot speed

$$s = \frac{f(2) - f(0)}{2 - 0} = 1.$$

Accordingly, the target solution is

$$u(x, t) = 2 \text{ for } x < t, \quad u(x, t) = 0 \text{ for } x > t.$$

This provides a benchmark in which the correct solution concept is explicit and the distinction between residual correctness and admissibility can be tested directly.

Because the target contains a discontinuity, ordinary pointwise error measures must be interpreted with care. A neural network cannot reproduce the jump exactly, and any approximation will necessarily replace the shock by a steep but continuous transition layer. For this reason, the later evaluation emphasizes not only global error but also shock location, shock compactness, overshoot or undershoot, and entropy-violation diagnostics. [3]

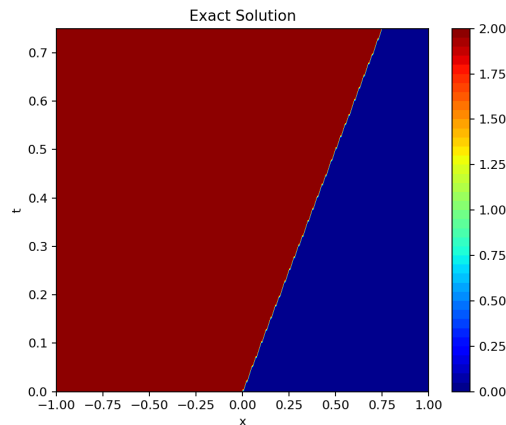


Figure 4: Entropy solution for the Burgers shock benchmark over the full space-time domain. The discontinuity travels along the line  $x = t$ , which provides the main reference structure for the hyperbolic comparisons.

## 6.4 Entropy-Aware Loss Construction

For Burgers' equation, the informed model uses the quadratic entropy

$$\eta(u) = \frac{u^2}{2}, \quad q(u) = \frac{u^3}{3}.$$

The corresponding entropy condition suggests penalizing the positive part of

$$\eta(u_\theta)_t + q(u_\theta)_x,$$

since admissible solutions should satisfy a nonpositive entropy production inequality. The resulting entropy-aware loss is then combined with the standard conservation-law residual and the soft initial-condition term. In this way, the network is trained not only to approximately satisfy the conservation law, but also to move toward the entropy solution class. [12]

## 6.5 Numerical Results

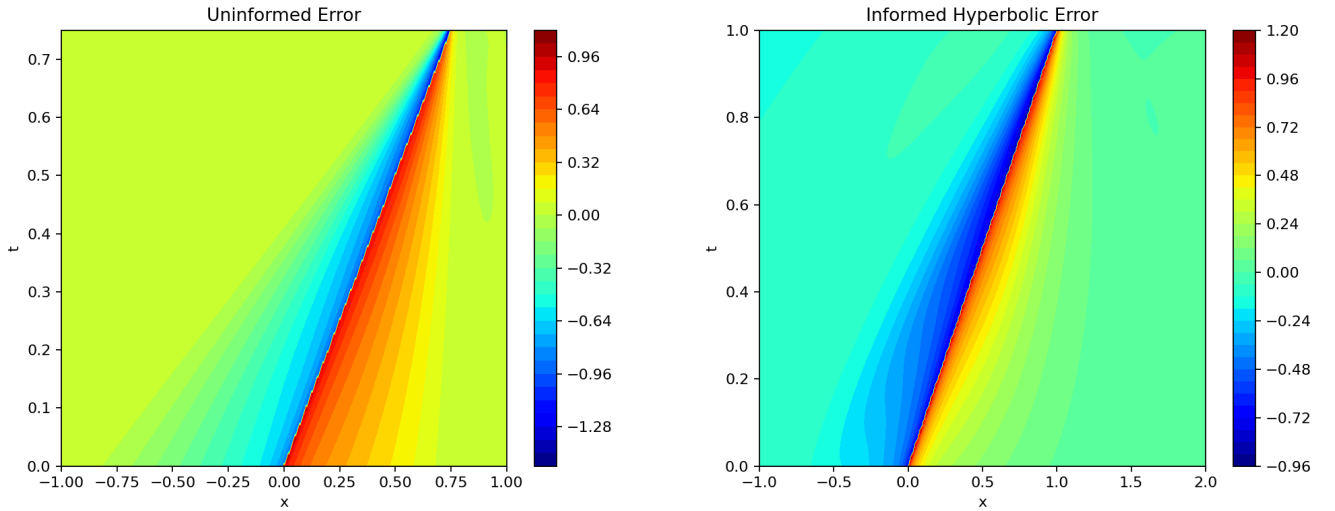


Figure 5: Error comparison for the hyperbolic shock benchmark. The uninformed model is shown on the left and the informed model is shown on the right. The left panel is a signed error plot and shows a broad positive/negative wedge around the shock, indicating a smeared and misaligned transition. The right panel is the absolute error for the informed model and shows that the remaining error is concentrated in a thin band along the shock trajectory.

The hyperbolic informed model achieved a relative  $L^1$  test error of approximately

$$1.57 \times 10^{-1},$$

with a final-time relative  $L^1$  error of approximately

$$7.04 \times 10^{-2}$$

and a final-time relative  $L^2$  error of approximately

$$8.96 \times 10^{-2}.$$

These values are substantially larger than in the parabolic and elliptic cases, but this is expected: the target is discontinuous, and the task is therefore intrinsically harder for a smooth neural ansatz.

On that criterion, the results are encouraging. The final shock location error is only about

$$2.5 \times 10^{-3},$$

indicating that the informed model places the shock at essentially the correct location. The solution also avoids overshoot entirely and exhibits only a very small undershoot of approximately

$$5.08 \times 10^{-3}.$$

These are important indicators of qualitative fidelity in a shock problem, where spurious oscillations and incorrect shock placement can be more informative than a single global norm.

The error plots make the qualitative improvement especially clear. In the uninformed model, the error is spread across a broad wedge-shaped region surrounding the shock. The signed error plot shows a clear negative band on one side of the shock and a positive band on the other, indicating that the network smears and misplaces the transition rather than sharply resolving it. By contrast, the informed model produces an absolute error field that is much more localized. Its error is concentrated in a thin narrow band along the shock trajectory  $x = t$ , with very little error away from the discontinuity.

One striking feature of the hyperbolic results is that a more informed model need not minimize every possible pointwise metric simultaneously. A sharper and more compact shock can sometimes lead to slightly larger local discrepancies near the discontinuity while still producing a better approximation of the entropy solution overall. For that reason, shock location, compactness, and entropy behavior are emphasized more heavily here than maximum pointwise error.

## 7 Elliptic Problems

Among the three PDE classes considered, the elliptic case is the one most naturally aligned with variational analysis. For coercive second-order elliptic equations, the standard theoretical framework is often built not from pointwise strong solutions, but from Sobolev spaces, weak formulations, and associated energy functionals. This makes the elliptic setting an especially natural place to move beyond a sampled strong-residual PINN and adopt a formulation that more closely matches the underlying PDE theory. [10, 11]

### 7.1 Dirichlet Trace Conditions and Admissible Classes

A central feature of elliptic boundary-value problems is the role of traces. In classical and weak theory alike, Dirichlet conditions are not merely auxiliary constraints added after the fact, but part of the definition of the admissible solution class. In Sobolev-space language, this is reflected in spaces such as  $H^1(\Omega)$ ,  $H_0^1(\Omega)$ , or affine trace classes determined by the prescribed boundary data. [10, 11]

This perspective motivates the informed elliptic neural ansatz used here. Rather than penalizing boundary mismatch softly, the trial space is built so that the boundary values are satisfied identically. This is analytically preferable because it removes one source of competition in the loss and makes the learned solution belong to the appropriate admissible class from the start. [8]

### 7.2 Variational Formulations and the Ritz Viewpoint

For coercive elliptic problems, variational structure often provides the most natural route to existence, uniqueness, and approximation. When the PDE can be written as the Euler–Lagrange equation of an energy functional, one may equivalently solve the problem by minimizing that functional over the admissible space. This is the classical Ritz viewpoint, and it is particularly well suited to neural trial functions because it replaces pointwise enforcement of the differential operator by optimization of a scalar functional over a flexible parameterized class. [10]

In the present elliptic case study, this is the main conceptual change relative to the uninformed baseline. The informed model still uses a neural network as the free approximation component, but training is organized around a discrete energy rather than a standard strong residual. This matters analytically because it changes the target notion of correctness. [6]

### 7.3 Maximum/Comparison-Principle Viewpoint

Although the variational formulation is the main analytical theme of this section, elliptic maximum and comparison principles still provide useful qualitative information. For suitable second-order elliptic operators, one may obtain interior extremum control and order-preservation statements from boundary data, forcing, and sign conditions on the coefficients. These principles are not the primary formulation used in the present elliptic model, but they motivate secondary diagnostics and soft barrier penalties that complement the variational loss. [13]

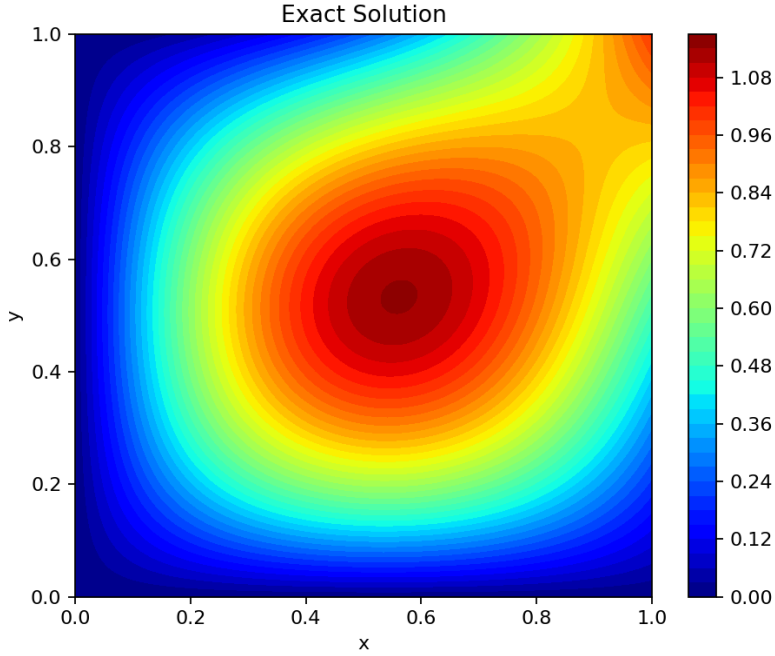


Figure 6: Exact solution for the elliptic benchmark on the square domain. The solution is smooth, nonnegative, and subject to nontrivial Dirichlet boundary data.

## 7.4 Benchmark Elliptical Problem

The elliptic benchmark is the variable-coefficient coercive Dirichlet problem

$$-\nabla \cdot (a(x, y) \nabla u) + c(x, y)u = f(x, y) \quad \text{in } \Omega = (0, 1)^2,$$

with exact solution

$$u(x, y) = \sin(\pi x) \sin(\pi y) + x^2 y.$$

The coefficients are chosen as

$$a(x, y) = 1 + x + y, \quad c(x, y) = 1 + x^2,$$

and the forcing  $f$  is defined accordingly so that the exact solution satisfies the equation. This benchmark is useful because it includes nontrivial variable coefficients and nonzero Dirichlet data on part of the boundary. [10, 11]

The solution is also nonnegative on  $[0, 1]^2$ , which makes it reasonable to supplement the variational training with a soft barrier-type qualitative constraint. [3]

## 7.5 Hard Boundary Enforcement

To satisfy the Dirichlet condition exactly, the informed model uses an admissible ansatz of the form

$$u_\theta(x, y) = g^*(x, y) + x(1 - x)y(1 - y) N_\theta(x, y),$$

where  $N_\theta$  is the free neural component and  $g^*$  is an extension of the prescribed boundary data. Because the factor  $x(1 - x)y(1 - y)$  vanishes on the entire boundary of the square, the neural correction disappears there and the boundary trace is fixed exactly by  $g^*$ . [8, 9]

This construction changes the admissible class of the approximation rather than merely penalizing trace mismatch. As a consequence, the boundary error is expected to be numerically negligible. [10, 11]

## 7.6 Variational Loss and Soft Max/Min-Style Barrier Loss

The informed elliptic model is trained using a discrete Ritz functional associated with the elliptic problem. Rather than minimizing a sampled strong residual, the network minimizes an approximation of the energy whose

Euler–Lagrange equation is the PDE itself. This makes the training objective much closer to the classical variational formulation. Alongside the variational loss, the model includes a soft barrier term that penalizes excursions outside a prescribed lower and upper range.

## 7.7 Numerical Results

The informed elliptic model produced a relative test error of approximately

$$1.51 \times 10^{-3},$$

together with a held-out residual mean square error of approximately

$$1.98 \times 10^{-3}.$$

More importantly for the elliptic setting, the boundary relative error is approximately

$$1.23 \times 10^{-7},$$

which is consistent with exact boundary enforcement up to numerical precision. The relative  $H^1$ -seminorm error is approximately

$$2.07 \times 10^{-3},$$

showing that the informed model is not only accurate in value but also in gradient structure, which is especially meaningful in a variational elliptic context.

The barrier diagnostics are similarly clean. The predicted solution remains in the expected admissible range, with

$$\text{min\_pred} = 0, \quad \text{max\_pred} \approx 1.1423,$$

and zero recorded lower, upper, or mean barrier violation on the evaluation grid.

The error plots make the improvement especially clear. The uninformed model exhibits a large signed-error pattern spread across the full domain. The error is not localized, but instead varies smoothly over the square, with the largest magnitudes occurring near the boundary and especially near the corners. By contrast, the informed model has a much smaller absolute error, with magnitude on the order of  $10^{-3}$  rather than  $10^{-1}$ . Its error field is smooth and concentrated in the interior of the domain, reaching its maximum near the center while decaying to essentially zero at the boundary.

This visual contrast is important for interpretation. The informed model does not merely reduce the error by a modest amount; it changes the qualitative structure of the error itself. The uninformed model shows a large-scale signed distortion across the solution, whereas the informed model removes the boundary mismatch almost completely and leaves only a small interior approximation error.

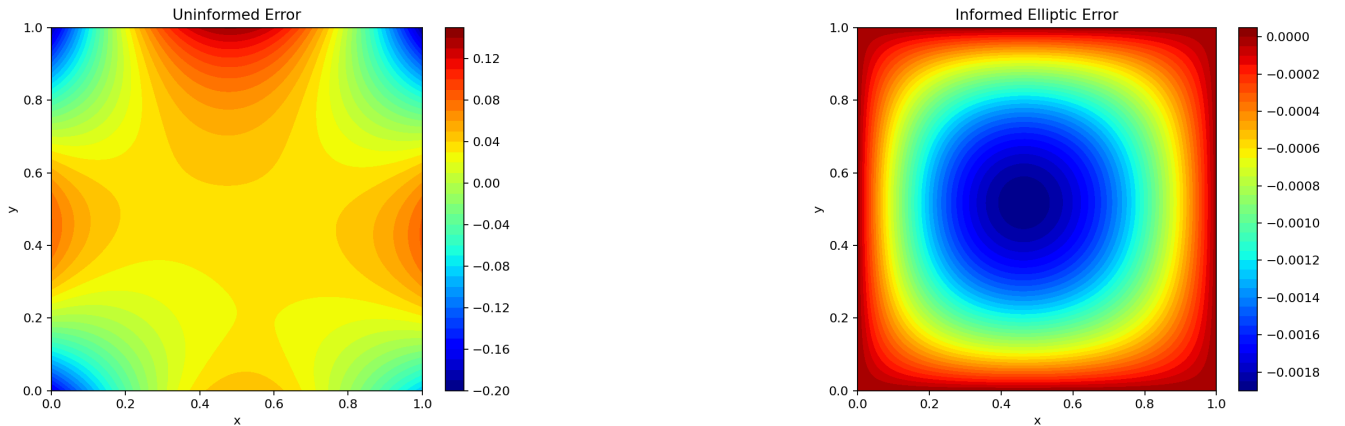


Figure 7: Error comparison for the elliptic benchmark. The uninformed model is shown on the left and the informed model is shown on the right. The left panel is a signed error plot and shows a broad, domain-wide error pattern with the largest magnitudes near the boundary and corners. The right panel is the absolute error for the informed model and shows a much smaller, smooth interior error that decays to essentially zero at the boundary.

## 8 Class Comparisons and Statistical Variability

The three case studies support a single overarching conclusion: correctness of a neural PDE solver is PDE-dependent. In the parabolic case, correctness is not exhausted by a small interior residual, but also includes accurate treatment of the initial surface, the spatial boundary, and comparison-principle-inspired qualitative behavior. In the hyperbolic case, correctness must include admissibility, since a small residual for the conservation law alone does not distinguish the entropy solution from other weak continuations after shock formation. In the elliptic case, correctness is most naturally expressed through admissible boundary traces and variational structure, with strong residuals playing a secondary diagnostic role rather than serving as the primary formulation. [3, 10, 11, 12]

These comparisons also motivate a brief statistical perspective. A trained neural PDE solver should not be viewed as a completely deterministic object, since its final output depends on random initialization, sampled collocation points, and the stochastic character of gradient-based optimization. In that sense, each trained model can be regarded as a random estimator produced by a learning procedure rather than as a single fixed numerical method. For this reason, repeated-seed summary statistics are useful even in a theory-first project: they quantify how reliably a given notion of correctness is reached in practice and help separate structural improvement from seed-specific accident. [14, 15, 3]

Table 1 summarizes the most important repeated-seed results for the three informed models using five independent runs in each case. The selected metrics are intentionally few and PDE-specific. For the parabolic model, the primary quantities are the relative solution error and final-time error, together with the mean barrier violation. For the hyperbolic model, relative  $L^1$  error, shock-location error, and mean entropy violation are more informative than a single pointwise metric. For the elliptic model, relative  $L^2$  and  $H^1$ -type errors are reported together with the boundary-trace error.

| PDE class  | Solution error                                     | Structure-aware metric 1                                       | Structure-aware metric 2                               |
|------------|----------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|
| Parabolic  | $\text{rel } L^2 = (2.70 \pm 1.30) \times 10^{-4}$ | $\text{final-time rel } L^2 = (2.73 \pm 1.06) \times 10^{-4}$  | $\text{mean barrier violation} = 5.85 \times 10^{-10}$ |
| Hyperbolic | $\text{rel } L^1 = 0.166 \pm 0.019$                | $\text{shock location error} = 0.015 \pm 0.015$                | $\text{mean entropy violation} = 0.0589 \pm 0.0080$    |
| Elliptic   | $\text{rel } L^2 = (1.43 \pm 0.63) \times 10^{-3}$ | $\text{rel } H^1\text{-semi} = (2.16 \pm 1.03) \times 10^{-3}$ | $\text{boundary trace error} = 1.23 \times 10^{-7}$    |

Table 1: Repeated-seed summary statistics for the three informed models, reported as mean  $\pm$  standard deviation over five runs when the variability is nontrivial. The metrics are chosen to reflect PDE-dependent notions of correctness rather than a single universal benchmark criterion.

Several conclusions follow immediately. The parabolic model is both highly accurate and highly stable, with essentially exact initial and boundary traces and negligible barrier violation. The elliptic model is similarly stable in its structural quantities, especially in boundary enforcement, while still maintaining small solution and gradient error under the variational formulation. The hyperbolic model is the least accurate in absolute terms, but this is the expected consequence of approximating a discontinuous entropy solution with a smooth neural ansatz. More importantly, its variability profile shows that the relevant hyperbolic quantities are shock location and entropy behavior rather than only a global norm.

## 9 Conclusions and Further Extensions

### 9.1 Main Conclusions

The central conclusion of this report is that correctness in neural PDE solvers is inherently PDE-dependent. A single strong-residual viewpoint does not capture the full mathematical structure relevant across parabolic, hyperbolic, and elliptic equations. In the parabolic case, correctness includes not only interior residual reduction but also exact treatment of initial and boundary traces together with qualitative comparison behavior. In the hyperbolic case, correctness must include admissibility, since weak satisfaction of the conservation law alone does not determine the entropy solution after shock formation. In the elliptic case, correctness is most naturally expressed through admissible boundary traces and variational structure, making Ritz-type training more faithful to the underlying PDE theory than naive pointwise residual minimization. [3, 10, 11, 12]

The computational results support this thesis. In each PDE class, the informed model improved on the uninformed baseline precisely because its loss and trial space were designed around the analytical features most natural for that class. This suggests that the most meaningful question is not whether a neural network can reduce a residual, but which mathematical statement the model is actually being trained to satisfy. [1, 2, 3]

## 9.2 Sparse-Data and Hybrid Data Extensions

A natural extension is to combine the present formulations with sparse observation data. Although this direction was not emphasized in the final comparisons, it remains relevant for partially observed systems, inverse problems, and settings in which exact solutions are unavailable but numerical or experimental samples can be obtained at selected points. In that case, the loss would contain both physics-informed terms and data-mismatch terms, leading to a hybrid formulation that interpolates between pure PINN training and data-driven surrogate learning. [1, 2]

This also raises questions of data augmentation and sampling design. Rather than treating observational points as uniformly distributed, one could study how performance changes when observations are concentrated near boundaries, initial surfaces, sharp fronts, or shock trajectories. Such sampling decisions are not merely computational details; they change how much information is supplied to the network and therefore interact directly with the role of the PDE constraints in training. [3]

## 9.3 Broader Uncertainty Quantification

The repeated-seed analysis in this report treated the trained neural solver as a random outcome of initialization, collocation sampling, and optimization. This empirical viewpoint is only a first step toward a broader uncertainty-quantification framework. One natural continuation would be to study probabilistic or Bayesian formulations in which uncertainty is attached directly to solution fields, parameters, or model discrepancy rather than inferred only from repeated training variability. Such approaches are especially relevant when observations are noisy, the PDE contains unknown parameters, or the model is used in inverse settings. [3, 16]

A second uncertainty direction is to study the statistical effect of the training design itself. Collocation points, barrier points, entropy points, and boundary samples are all chosen numerically, and different sampling strategies may induce different variability profiles. This suggests a more explicitly statistical view in which the learned solution is understood not only through its deterministic loss but also through the sampling distribution and optimization procedure that produced it. In that sense, uncertainty analysis for PINNs includes both classical posterior-style questions and empirical robustness questions tied to stochastic training. [14, 15, 3]

## 9.4 Further PDE Classes and Formulations

Several additional extensions follow naturally from the present work. One direction is to enlarge the set of qualitative constraints encoded in the loss or architecture. Beyond the barrier and entropy constraints used here, one could study monotonicity, convexity, conservation, positivity, or other structural properties suggested by PDE theory. Another is to replace soft penalties by more explicit constrained-optimization mechanisms, such as Lagrange-multiplier or augmented-Lagrangian formulations, which may offer a more principled treatment of competing constraints than fixed penalty weights alone. [3]

A second direction is to vary the PDE formulation more aggressively. In the hyperbolic setting, first-order PINNs provide an alternative route for conservation laws by reformulating the problem in terms of lower-order derivative structure, which may be advantageous near shocks and discontinuities. More broadly, one could study weak-form or domain-decomposed formulations beyond the single informed models considered here. [17, 6, 7]

A third direction concerns model architecture. The present report used simple coordinate-based multilayer perceptrons in order to isolate the effect of constraint encoding. Future work could instead examine architectures tailored to additional structure, including Lagrangian or Hamiltonian neural networks for mechanics-inspired systems, or more global operator-learning architectures such as Fourier neural operators and DeepONet-type models. These are especially relevant when the goal shifts from solving a single PDE instance to learning families of PDE solution maps or full operators between function spaces. [4, 5, 14]

Finally, one may broaden the scope of the PDE classes themselves. The present study treated one representative benchmark from each of the parabolic, hyperbolic, and elliptic categories. A natural next step would be to examine broader families of coefficients, geometries, nonlinearities, and boundary conditions, and to study whether the same conclusions persist under more general formulations. In that expanded setting, the main theme of the report would remain unchanged: the success of an informed neural PDE solver should be judged by how well its encoded constraints align with the analytical structure of the PDE class under consideration. [10, 11, 12, 3]



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